The sign of the mass self-energy in the Born analysis of "Anomalous quantum Hall effect induced by disorder in topological insulators"

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Abstract

The self-consistent Born approximation (SCBA) used in Ref. [1], and in Chapter 2 of the thesis [2], renormalises the masses of the two spin blocks with the wrong sign. The error is in the definition of the renormalised (barred) parameters, $\mathrm{Re}\,\Sigma \equiv -\mathrm{diag}[(\bar{m}-m)\tau_z + (\bar{\mu}-\mu)\tau_0, \dots]$, which holds for the chemical potential but not for the mass. The program that computed the Born phase diagrams (Fig. 4 of Ref. [1]) implements these equations as written. With the correct sign, the disorder fluctuations push both masses towards the topological side, as in the topological Anderson insulator that the paper follows. Three checks confirm it, none of which uses the derivation: a full matrix SCBA, the exact disorder-averaged Green's function, and spin-resolved transport. The numerical results of the paper (Figs. 1–3) are unaffected, and so is its main conclusion: a $G = 1$ Chern-insulator phase, carried by the spin-down edge state, for both couplings. What changes is the Born interpretation. In the same-g ("ferromagnetic") case, the spin-down block, which carries the $G = 1$ plateau, never changes sign, whereas the published lines invert it. The spin-up channel is lost because the Fermi level enters the spin-up band, shifted by the mean exchange field. At stronger disorder an impurity band of the spin-up block crosses the Fermi level, a regime beyond any Born approximation, and a trivial gap opens only after the spin-up channel is lost. In the opposite-g ("antiferromagnetic") case, the spin-up mass changes sign at $J_I \simeq 0.40$, not 0.30. The apparent agreement of the published inversion line with the antiferromagnetic conductance map is a finite-width effect of the 64-site ribbons.

Contents

1 Summary **2**

2 The model and the published equations **3**

3 The self-consistent Born approximation, step by step **4**

3.1 Step 1: the clean model in $\boldsymbol{k}$ space 4
3.2 Step 2: the disorder and its first two moments 4
3.3 Step 3: averaged Green's function, mean field and self-energy 5
3.4 Step 4: second order, the Born approximation 5
3.5 Step 5: the vertex algebra 6
3.6 Step 6: the local Green's function of a block 6
3.7 Step 7: the renormalised parameters 6
3.8 Step 8: the four self-consistent equations 6
3.9 Step 9: band edges and topology, the lines of Fig. 4 7
3.10 Step 10: lowest order, the logarithm and the sign of the mass correction 7
3.11 Step 11: the numerical solution in the program 7

4 The error **8**
4.1 Derivation . . . 8
4.2 One-line check . . . 8
4.3 It is not a sign convention . . . 8
4.4 Consistency with the perturbative result and with the topological Anderson insulator 8
4.5 A second correction: the longitudinal weight . . . 9

5 Checks that do not use the derivation **9**
5.1 Matrix SCBA . . . 9
5.2 Exact renormalised masses . . . 9
5.3 Spin-resolved transport . . . 13

6 Impact on the Born phase diagrams **13**

7 Why the published inversion line matched the antiferromagnetic map **17**
7.1 Overlap of the edge states . . . 17
7.2 Wider ribbons . . . 17
7.3 The coincidence . . . 18

8 Origin and stability of the $G = 1$ phase **19**

9 What changes for the paper **21**

A Methods and reproducibility **22**

1 Summary

- **The derivation** (Sec. 3). Every step of the SCBA, from the Hamiltonian to the program; the error is in Step 7.
- **The error.** The mass equations of the SCBA carry the wrong sign (Sec. 4). The chemical-potential equations are right, as the exact $\bar{\mu}$ confirm (Sec. 5.2). The weight v_z of the longitudinal fluctuations is also too large, by a factor $1 + n_I$; this correction alone moves the opposite-g inversion by 0.06 (Sec. 4.5).
- **Where it comes from.** The derivation, as printed in Ref. [1] and in the thesis [2]: the code `self_born.f90` implements the printed equations faithfully. A replica of the program reproduces the published Born data (runs 00–07) to 10^{-14}.
- **Independent evidence** (Sec. 5):
  1. a matrix SCBA computed from the Hamiltonian, without the reduction;
  2. the exact renormalised masses, from the disorder-averaged Green's function (Fig. 1);
  3. the spin-resolved conductance of the same ribbons as the paper.

  All three follow the corrected sign. At weak disorder the corrected SCBA agrees with the exact masses to 10^{-3}, while the published one moves them in the opposite direction.
- **Impact on Fig. 4.** With the published maps (Fig. 3, Sec. 6):
  - same-g: the corrected lines predict $G = 1$ on 99.9 % of the $G = 1$ plateau, the published ones on 20 %. This is coverage, not precision: the corrected lines also predict $G = 1$ on the crossover before the plateau, so only 35 % of the points where they predict it lie on the plateau (17 % for the published lines, 26 % for a prediction of $G = 1$ everywhere);

– opposite-g: the published spin-up inversion line sits at the left edge of the $G = 1$ region, the corrected one inside it.

- **Why the published lines looked right** (Sec. 7). The ribbons are 64 sites wide. Close to the inversion, the spin-up edge states decay over $1.5/|\bar{m}_\uparrow|$ sites, so they overlap across the ribbon and the spin-up channel is lost before the mass changes sign. Wider ribbons move the transition towards the exact inversion. At $W = 256$ the spin-up block still carries $T_\uparrow = 0.99$ where the published masses make it trivial. The wrong sign moves the Born inversion by about the same amount in the same direction: a coincidence.

- **Origin and stability of** $G = 1$ (Sec. 8). $G = 1$ is a Chern insulator: the spin-down block keeps its topological gap, while the spin-up states at the Fermi level no longer conduct. Each edge then carries a single chiral mode, with nothing to backscatter into. The same-g plateau does not depend on the ribbon width, and it resists tilted moments ($M_z = 0.8$) better than $G = 2$: it keeps 13.9 % of the map (26 % at $M_z = 1$), while $G = 2$ falls from 23 % to 3.8 %.

2 The model and the published equations

The two spin blocks of the BHZ model are, on the square lattice (lattice constant 1),

$$h_\sigma(\boldsymbol{k}) = -D_{\boldsymbol{k}}\,\tau_0 + (m - B_{\boldsymbol{k}})\,\tau_z + A\,(\text{terms odd in } \boldsymbol{k}), \qquad B_{\boldsymbol{k}} = B(4 - 2\cos k_x - 2\cos k_y), \quad (1)$$

and similarly for $D_{\boldsymbol{k}}$. The parameters are $A = 1$, $B = -0.376406$, $D = -0.280933$ and $m = -0.137$, so the clean system is topological, $m/B > 0$. The Pauli matrices τ act on the electron (E) and hole (H) orbitals.

The magnetic impurities occupy a fraction n_I of the sites. They carry classical moments $\boldsymbol{S}_i$ with $\langle S_z\rangle = M_z$ and couple through

$$V_\alpha = J_I \sum_{i\in I} \boldsymbol{S}_i \cdot \boldsymbol{\sigma}\,\tau_\alpha, \qquad \tau_\alpha = \tau_0 \text{ (same-}g\text{, “F”)}, \quad \tau_\alpha = \tau_z \text{ (opposite-}g\text{, “AF”)}. \quad (2)$$

This note treats the fully polarised case $M_z = 1$ of the paper's Figs. 1, 2 and 4. Then $V_\alpha = J_I\sigma_z\tau_\alpha$ on the impurity sites, s_z is conserved, and the two spin blocks are independent disordered systems. The mean field $\langle V_\alpha\rangle = n_I J_I M_z\,\sigma_z\tau_\alpha$ shifts, block by block:

- the chemical potential, $\mu_\sigma = \mu - \sigma n_I J_I M_z$, for same-$g$;
- the mass, $m_\sigma = m + \sigma n_I J_I M_z$, for opposite-$g$.

Ref. [1] (Secs. IV A “Born approximation” and IV B “Phase diagram”) and the thesis [2] (Sec. “Discussion : l'approximation de Born”) write the mean-field Green's function and the SCBA self-energy as

$$g_\alpha(E,\boldsymbol{k}) = [E + \mu - H_0(\boldsymbol{k}) - \langle V_\alpha\rangle]^{-1}, \qquad \Sigma(E) = \frac{n_I J_I^2}{N^2}\sum_{\boldsymbol{k}} [v_z\, g_z + v_\perp g_\perp], \quad (3)$$

where g is replaced by the full averaged Green's function in the self-consistent version. They then *define* the renormalised parameters by

$$\mathrm{Re}\,\Sigma \equiv -\mathrm{diag}\big[(\bar{m}_\uparrow - m_\uparrow)\tau_z + (\bar{\mu}_\uparrow - \mu_\uparrow)\tau_0,\ (\bar{m}_\downarrow - m_\downarrow)\tau_z - (\bar{\mu}_\downarrow - \mu_\downarrow)\tau_0\big] \qquad \text{(as printed)}, \quad (4)$$

and obtain

$$\bar{\mu}_\sigma = \mu_\sigma - \frac{n_I J_I^2}{N^2}\sum_{\boldsymbol{k}}\Big[\frac{v_z}{\Delta_\sigma}(E_F + \bar{\mu}_\sigma + D_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar\sigma}}(E_F + \bar{\mu}_{\bar\sigma} + D_{\boldsymbol{k}})\Big], \quad (5)$$

$$\bar{m}_\sigma = m_\sigma - \frac{n_I J_I^2}{N^2}\sum_{\boldsymbol{k}}\Big[\frac{v_z}{\Delta_\sigma}(\bar{m}_\sigma - B_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar\sigma}}(\bar{m}_{\bar\sigma} - B_{\boldsymbol{k}})\Big], \quad (6)$$

with $\Delta_\sigma = (E_F + \bar{\mu}_\sigma + D_{\boldsymbol{k}})^2 - (\bar{m}_\sigma - B_{\boldsymbol{k}})^2 - \sin^2 k_x - \sin^2 k_y$. The gap of block σ is centred on $-\bar{\mu}_\sigma$, with half-width $|\bar{m}_\sigma|$, and the block is topological when $\bar{m}_\sigma/B > 0$. Fig. 4 of Ref. [1] draws the band edges $E_F = -\bar{\mu}_\sigma \pm \bar{m}_\sigma$ and the lines $\bar{m}_\sigma = 0$.[1]

The program `self_born.f90`, which produced Fig. 4, iterates exactly Eqs. (5) and (6). In it, `norm` is $n_I J_I^2/N^2$, `kc` is $B_{\boldsymbol{k}}/B$, and `vzu = vz/detu` with `detu` $= \Delta_\uparrow$:

```
154  vz = (1-conc**2)*Mz**2 + (1-Mz)**2/3.0
192  deltaMu = -norm * sum( vzu*(Mu - B*kc) + vperpd*(Md - B*kc) )
200  Mu_temp = Mu0 + deltaMu
```

A replica written for this note reproduces the published Born data of runs 00–07 to 10^{-14}. The code is therefore a faithful implementation of the printed equations.

3 The self-consistent Born approximation, step by step

This section rederives the SCBA of Ref. [1] from the Hamiltonian, with every intermediate step, in the paper's notation. It departs from the printed equations in two places: the identification of the renormalised mass (Step 7) and the weight v_z (Step 2). Both are flagged where they occur. Spins are labelled $\sigma = +1$ ($\uparrow$) and $\sigma = -1$ ($\downarrow$), and $\bar{\sigma} = -\sigma$.

3.1 Step 1: the clean model in $\boldsymbol{k}$ space

Each site carries four orbitals, $(E\uparrow, H\uparrow, E\downarrow, H\downarrow)$. The Pauli matrices $\boldsymbol{\sigma}$ act on the spin and $\boldsymbol{\tau}$ on the E/H orbitals. After a Fourier transform on the square lattice, the clean Hamiltonian is block diagonal in spin, $H_0(\boldsymbol{k}) = \mathrm{diag}[h_\uparrow(\boldsymbol{k}), h_\downarrow(\boldsymbol{k})]$, with

$$h_\sigma(\boldsymbol{k}) = -D_{\boldsymbol{k}}\,\tau_0 + (m - B_{\boldsymbol{k}})\,\tau_z + A\,(\sigma \sin k_x\,\tau_x - \sin k_y\,\tau_y), \qquad B_{\boldsymbol{k}} = B\,(4 - 2\cos k_x - 2\cos k_y), \quad (7)$$

and the same for $D_{\boldsymbol{k}}$. Two properties of the τ_x, τ_y terms are used below:

- they are odd in $\boldsymbol{k}$;
- their squares add up to $A^2(\sin^2 k_x + \sin^2 k_y)$.

The chemical potential enters as $H_0 - \mu$, so Green's functions are $[E + \mu - H]^{-1}$. In this form the mass appears with a plus sign ($+m\,\tau_z$) and the chemical potential with a minus sign ($-\mu\,\tau_0$). This difference is what Step 7 is about.

3.2 Step 2: the disorder and its first two moments

Each site $\boldsymbol{r}$ is occupied by an impurity with probability n_I, independently of the other sites: $\eta_{\boldsymbol{r}} \in \{0, 1\}$, $\langle \eta_{\boldsymbol{r}} \rangle = n_I$. An impurity carries a classical unit moment $\boldsymbol{S}_{\boldsymbol{r}}$, independent of η, with

- a uniform azimuth, so $\langle S_x \rangle = \langle S_y \rangle = 0$ and $\langle S_a S_b \rangle = 0$ for $a \neq b$;
- S_z uniform in $[2M_z - 1, 1]$ (the polarised moments of the thesis), so $\langle S_z \rangle = M_z$ and $\langle S_z^2 \rangle = M_z^2 + (1 - M_z)^2/3$;
- $\langle S_x^2 \rangle = \langle S_y^2 \rangle = (1 - \langle S_z^2 \rangle)/2$, since $|\boldsymbol{S}| = 1$.

[1] Eq. (4) as printed also has opposite signs for the τ_0 terms of the two blocks. Eqs. (5) use the same sign for both, which is the right one. The thesis writes $\mu_\uparrow$ instead of $m_\uparrow$ on the right-hand side of the spin-up mass equation; the program uses $m_\uparrow$.

The potential on site $\boldsymbol{r}$ is

$$U_{\boldsymbol{r}} = J_I\,\eta_{\boldsymbol{r}}\,(\boldsymbol{S}_{\boldsymbol{r}}\cdot\boldsymbol{\sigma})\,\tau_\alpha = J_I\sum_{a=x,y,z} X^{\boldsymbol{r}}_a\,\sigma_a\tau_\alpha, \qquad X_a = \eta\,S_a. \tag{8}$$

Its mean is $\langle X_a\rangle = n_I\langle S_a\rangle$, so

$$\langle U\rangle = n_I J_I M_z\,\sigma_z\tau_\alpha. \tag{9}$$

Its covariances follow from $\eta^2=\eta$: $C_{ab} = \langle X_aX_b\rangle - \langle X_a\rangle\langle X_b\rangle = n_I\langle S_aS_b\rangle - n_I^2\langle S_a\rangle\langle S_b\rangle$. Only the diagonal ones survive:

$$C_{zz} = n_I(\langle S_z^2\rangle - n_I M_z^2) \equiv n_I\,v_z, \qquad C_{xx}=C_{yy} = \tfrac{n_I}{2}(1-\langle S_z^2\rangle) \equiv \tfrac{n_I}{2}\,v_\perp, \tag{10}$$

$$v_z = (1-n_I)M_z^2 + \tfrac{1}{3}(1-M_z)^2, \qquad v_\perp = \tfrac{2}{3}(1-M_z)(1+2M_z). \tag{11}$$

Ref. [1] writes $v_z = \langle S_z^2\rangle - n_I^2M_z^2$: the factor n_I is taken out twice (Sec. 4.5). $v_\perp$ agrees.

3.3 Step 3: averaged Green's function, mean field and self-energy

For one configuration, $G=[z+\mu-H_0-V]^{-1}$, with $z=E+\mathrm{i}0^+$ and $V=\sum_{\boldsymbol{r}}U_{\boldsymbol{r}}$. The disorder average restores translation invariance, so $\langle G\rangle$ is diagonal in $\boldsymbol{k}$. The self-energy is defined by the Dyson equation:

$$\langle G\rangle(\boldsymbol{k}) = \big[z+\mu-H_0(\boldsymbol{k})-\langle V\rangle-\Sigma(\boldsymbol{k},z)\big]^{-1}. \tag{12}$$

The mean field $\langle V\rangle=\langle U\rangle$ is uniform. As in the paper, it goes into the unperturbed Green's function, $g(\boldsymbol{k})=[z+\mu-H_0(\boldsymbol{k})-\langle V\rangle]^{-1}$. Block by block:

- same-g ($\tau_\alpha=\tau_0$): $\langle V\rangle_\sigma = \sigma n_I J_I M_z\,\tau_0$, so $\mu_\sigma=\mu-\sigma n_I J_I M_z$ and $m_\sigma=m$;
- opposite-g ($\tau_\alpha=\tau_z$): $\langle V\rangle_\sigma = \sigma n_I J_I M_z\,\tau_z$, so $\mu_\sigma=\mu$ and $m_\sigma = m+\sigma n_I J_I M_z$.

Then $g_\sigma(\boldsymbol{k})=[z+\mu_\sigma-h_\sigma(\boldsymbol{k};m_\sigma)]^{-1}$, where $h_\sigma(\boldsymbol{k};m_\sigma)$ is Eq. (7) with m replaced by m_σ.

3.4 Step 4: second order, the Born approximation

Write $\delta U_{\boldsymbol{r}} = U_{\boldsymbol{r}} - \langle U\rangle$ and expand: $G = g + g\,\delta V g + g\,\delta V g\,\delta V g + \ldots$ Since $\langle\delta V\rangle=0$,

$$\langle G\rangle = g + g\,\langle \delta V g\,\delta V\rangle\,g + O(\delta V^3) \quad\Longrightarrow\quad \Sigma = \langle \delta V g\,\delta V\rangle + O(\delta V^3). \tag{13}$$

On the sites, $(\delta V g\,\delta V)_{\boldsymbol{r}\boldsymbol{r}'} = \delta U_{\boldsymbol{r}}\,g_{\boldsymbol{r}\boldsymbol{r}'}\,\delta U_{\boldsymbol{r}'}$. Different sites are independent and δU has zero mean, so the average vanishes unless $\boldsymbol{r}=\boldsymbol{r}'$:

$$\Sigma_{\boldsymbol{r}\boldsymbol{r}'} = \delta_{\boldsymbol{r}\boldsymbol{r}'}\,\langle \delta U\,g_{\mathrm{loc}}\,\delta U\rangle, \qquad g_{\mathrm{loc}} = g_{\boldsymbol{r}\boldsymbol{r}} = \frac{1}{N_k}\sum_{\boldsymbol{k}} g(\boldsymbol{k}). \tag{14}$$

The self-energy is local, hence independent of $\boldsymbol{k}$. N_k is the number of $\boldsymbol{k}$ points (N^2 in the paper). The self-consistent version replaces g by $\langle G\rangle$ itself, which resums all the non-crossing diagrams:

$$\Sigma = \langle \delta U\,\langle G\rangle_{\mathrm{loc}}\,\delta U\rangle, \qquad \langle G\rangle_{\mathrm{loc}} = \frac{1}{N_k}\sum_{\boldsymbol{k}}\big[z+\mu-H_0(\boldsymbol{k})-\langle V\rangle-\Sigma\big]^{-1}. \tag{15}$$

3.5 Step 5: the vertex algebra

With the covariances of Step 2,

$$\Sigma = J_I^2 \sum_{ab} C_{ab}\,(\sigma_a\tau_\alpha)\langle G\rangle_{\rm loc}(\sigma_b\tau_\alpha) = n_I J_I^2\Big[v_z\,(\sigma_z\tau_\alpha)\langle G\rangle_{\rm loc}(\sigma_z\tau_\alpha) + \frac{v_\perp}{2}\sum_{a=x,y}(\sigma_a\tau_\alpha)\langle G\rangle_{\rm loc}(\sigma_a\tau_\alpha)\Big]. \tag{16}$$

Take $\langle G\rangle$ block diagonal in spin; Eq. (15) then keeps it so. For a block-diagonal matrix,

$$\sigma_z\,\mathrm{diag}(P,Q)\,\sigma_z = \mathrm{diag}(P,Q), \qquad \sigma_x\,\mathrm{diag}(P,Q)\,\sigma_x = \sigma_y\,\mathrm{diag}(P,Q)\,\sigma_y = \mathrm{diag}(Q,P). \tag{17}$$

Longitudinal fluctuations keep the spin and transverse ones exchange the blocks:

$$\Sigma_\sigma = n_I J_I^2\big[v_z\,\tau_\alpha\langle G_\sigma\rangle_{\rm loc}\tau_\alpha + v_\perp\,\tau_\alpha\langle G_{\bar\sigma}\rangle_{\rm loc}\tau_\alpha\big]. \tag{18}$$

Step 6 shows that $\langle G_\sigma\rangle_{\rm loc}$ has only τ_0 and τ_z components. These commute with τ_z, so τ_α drops out for both couplings:

$$\Sigma_\sigma = n_I J_I^2\big[v_z\,\langle G_\sigma\rangle_{\rm loc} + v_\perp\,\langle G_{\bar\sigma}\rangle_{\rm loc}\big]. \tag{19}$$

At this order the two couplings differ only through the mean field of Step 3.

3.6 Step 6: the local Green's function of a block

By Eq. (19), $\Sigma_\sigma = \Sigma^0_\sigma\tau_0 + \Sigma^z_\sigma\tau_z$. The averaged block is then

$$\langle G_\sigma\rangle(\boldsymbol{k}) = \big[a_\sigma(\boldsymbol{k}) - \boldsymbol{d}_\sigma(\boldsymbol{k})\cdot\boldsymbol{\tau}\big]^{-1}, \qquad a_\sigma = z + \mu_\sigma + D_{\boldsymbol{k}} - \Sigma^0_\sigma, \quad d^z_\sigma = m_\sigma - B_{\boldsymbol{k}} + \Sigma^z_\sigma, \tag{20}$$

with $d^x_\sigma = \sigma A\sin k_x$ and $d^y_\sigma = -A\sin k_y$. Since $(\boldsymbol{d}\cdot\boldsymbol{\tau})^2 = \boldsymbol{d}\cdot\boldsymbol{d}$,

$$\big[a - \boldsymbol{d}\cdot\boldsymbol{\tau}\big]^{-1} = \frac{a + \boldsymbol{d}\cdot\boldsymbol{\tau}}{a^2 - \boldsymbol{d}\cdot\boldsymbol{d}}, \qquad \Delta_\sigma(\boldsymbol{k}) = a_\sigma^2 - \boldsymbol{d}_\sigma\cdot\boldsymbol{d}_\sigma. \tag{21}$$

The sum over $\boldsymbol{k}$ removes the odd components d^x, d^y:

$$\langle G_\sigma\rangle_{\rm loc} = \frac{1}{N_k}\sum_{\boldsymbol{k}}\frac{a_\sigma(\boldsymbol{k})\,\tau_0 + d^z_\sigma(\boldsymbol{k})\,\tau_z}{\Delta_\sigma(\boldsymbol{k})}. \tag{22}$$

3.7 Step 7: the renormalised parameters

The barred parameters are defined so that the averaged block is a clean block with new parameters, $a_\sigma = z + \bar\mu_\sigma + D_{\boldsymbol{k}}$ and $d^z_\sigma = \bar m_\sigma - B_{\boldsymbol{k}}$:

$$\bar\mu_\sigma = \mu_\sigma - \Sigma^0_\sigma, \qquad \bar m_\sigma = m_\sigma + \Sigma^z_\sigma, \qquad \text{i.e.} \qquad \mathrm{Re}\,\Sigma_\sigma = (\bar m_\sigma - m_\sigma)\,\tau_z - (\bar\mu_\sigma - \mu_\sigma)\,\tau_0. \tag{23}$$

This is where the printed equations go wrong. Eq. (4) puts a minus sign in front of both terms, i.e. $\bar m_\sigma = m_\sigma - \Sigma^z_\sigma$. The chemical potential enters $z + \mu - h$ with a plus sign, and the mass enters through $-h$, i.e. with a minus sign. Their self-energy corrections therefore have opposite signs. With the barred parameters, $\Delta_\sigma = (z + \bar\mu_\sigma + D_{\boldsymbol{k}})^2 - (\bar m_\sigma - B_{\boldsymbol{k}})^2 - A^2(\sin^2 k_x + \sin^2 k_y)$, the paper's determinant.

3.8 Step 8: the four self-consistent equations

Inserting Eq. (22) into Eq. (19), component by component:

$$\Sigma^0_\sigma = \frac{n_I J_I^2}{N_k}\sum_{\boldsymbol{k}}\Big[\frac{v_z}{\Delta_\sigma}(z + \bar\mu_\sigma + D_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar\sigma}}(z + \bar\mu_{\bar\sigma} + D_{\boldsymbol{k}})\Big], \tag{24}$$

$$\Sigma^z_\sigma = \frac{n_I J_I^2}{N_k}\sum_{\boldsymbol{k}}\Big[\frac{v_z}{\Delta_\sigma}(\bar m_\sigma - B_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar\sigma}}(\bar m_{\bar\sigma} - B_{\boldsymbol{k}})\Big]. \tag{25}$$

Then $\bar\mu_\sigma = \mu_\sigma - \Sigma^0_\sigma$, which is Eq. (5) as printed, and $\bar m_\sigma = m_\sigma + \Sigma^z_\sigma$, which is Eq. (31) and differs from the printed Eq. (6) by the sign. These are four equations for the four complex unknowns $(\bar\mu_\uparrow, \bar\mu_\downarrow, \bar m_\uparrow, \bar m_\downarrow)$ at each (J_I, E_F), with $z = E_F + \mathrm{i}\eta$. For $M_z = 1$, $v_\perp = 0$ and the two blocks decouple.

3.9 Step 9: band edges and topology, the lines of Fig. 4

The renormalised block has its gap at $\boldsymbol{k} = 0$, where $D_{\boldsymbol{k}} = B_{\boldsymbol{k}} = \sin k = 0$: its bands start at $E = -\bar{\mu}_\sigma \pm \bar{m}_\sigma$ (real parts). Hence:

- the band edges are $E_F = -\bar{\mu}_\sigma \pm \bar{m}_\sigma$. Since the parameters depend on E_F through z, each line of Fig. 4 is the zero contour of $E_F + \bar{\mu}_\sigma \mp \bar{m}_\sigma$ on the (J_I, E_F) grid;
- E_F lies in the gap of block σ when $|E_F + \bar{\mu}_\sigma| < |\bar{m}_\sigma|$;
- block σ is topological when $\bar{m}_\sigma/B > 0$, a band inversion at Γ (on the lattice, for $0 < \bar{m}_\sigma/B < 4$). The line $\bar{m}_\sigma = 0$ is its topological transition.

3.10 Step 10: lowest order, the logarithm and the sign of the mass correction

Put the mean-field parameters on the right-hand side of Step 8 and go to the continuum, $N_k^{-1} \sum_{\boldsymbol{k}} \to \int_{|\boldsymbol{k}|<k_*} d^2k/(2\pi)^2$ with $k_* = \pi$:

$$\Sigma^z_\sigma \simeq n_I J_I^2 v_z \int \frac{d^2k}{(2\pi)^2} \frac{m_\sigma - Bk^2}{\Delta_\sigma(\boldsymbol{k})}, \qquad \Delta_\sigma(\boldsymbol{k}) = (E_F + \mu_\sigma + Dk^2)^2 - (m_\sigma - Bk^2)^2 - A^2k^2. \quad (26)$$

At large k, $\Delta_\sigma \simeq -(B^2 - D^2)k^4$ and the integrand tends to $B/[(B^2 - D^2)k^2]$. Its integral is logarithmic:

$$\int_{k_0}^{k_*} \frac{k\,dk}{2\pi} \frac{B}{(B^2 - D^2)k^2} = \frac{B}{2\pi(B^2 - D^2)} \ln \frac{k_*}{k_0}. \quad (27)$$

It is cut off at small k, where $(B^2 - D^2)k_0^4 \simeq |\Delta_\sigma(E_F, 0)|$. Hence

$$\mathrm{Re}\,\Sigma^z_\sigma \simeq \frac{n_I J_I^2 v_z}{8\pi} \frac{B}{B^2 - D^2} \ln \Big| \frac{(B^2 - D^2)k_*^4}{\Delta_\sigma(E_F, 0)} \Big|, \quad (28)$$

and in the same way $\mathrm{Re}\,\Sigma^0_\sigma \simeq -\frac{n_I J_I^2 v_z}{8\pi} \frac{D}{B^2 - D^2} \ln |\dots|$. This is Eq. (32) of Sec. 4, the paper's lowest-order result.

Sign. $B^2 - D^2 > 0$, and the logarithm is large and positive when E_F is in the gap, so $\mathrm{Re}\,\Sigma^z$ has the sign of B. With Step 7, $\bar{m}_\sigma - m_\sigma = \mathrm{Re}\,\Sigma^z_\sigma$: the mass moves away from zero on the topological side. With the printed sign, it moves towards the trivial side.

Numbers. Take same-g, $n_I = 0.2$, $J_I = 0.3$, $E_F = 0$ and spin up, so $\mu_\uparrow = -0.06$. Then $B/(B^2 - D^2) = -6.0$, the logarithm is 6.0 and, with the corrected weight $v_z = 0.8$, $\mathrm{Re}\,\Sigma^z \simeq -0.021$.

- Corrected sign: $\bar{m}_\uparrow \simeq -0.158$, against -0.149 for the full corrected SCBA and -0.153 for the exact mass (Table 3).
- Printed sign: $\bar{m}_\uparrow \simeq -0.116$. With the printed weight as well, $v_z = 0.96$ and $\mathrm{Re}\,\Sigma^z \simeq -0.025$, it is -0.112, against -0.123 for the published SCBA.

3.11 Step 11: the numerical solution in the program

`self_born.f90` solves Step 8 at each point of a 64×64 grid of (J_I, E_F):

- the $\boldsymbol{k}$ sum runs over a 256×64 grid of the Brillouin zone, with `norm` $= n_I J_I^2/(256 \cdot 64)$, `kc` $= 4 - 2\cos k_x - 2\cos k_y$ and `ks` $= \sin^2 k_x + \sin^2 k_y$;
- $z = E_F + \mathrm{i}\eta$ with $\eta = 10^{-3}$;

- the unknowns are `Mu`, `Md` ($\bar m_\uparrow$, $\bar m_\downarrow$) and `Muu`, `Mud` ($\bar\mu_\uparrow$, $\bar\mu_\downarrow$). They start from the mean field, `Mu0 = mass + anti*J_I*conc*Mz` and `Muu0 = -(1-anti)*J_I*conc*Mz`, with `anti` $= 1$ for opposite-g;
- the iteration is a damped fixed point, $X \leftarrow pF(X) + (1-p)X$: first with $p = 0.15$ down to a tolerance 10^{-4} (at most 500 iterations), then with $p = 0.8$ down to 10^{-6};
- the program writes the real parts of $\bar m_\sigma$ and $\bar\mu_\sigma$.

The replica in `caroli` (`ScbaAnalytic`) reproduces it to 10^{-14}. `ScbaAnalytic::point_corrected` changes only the sign of Step 7 and the weight of Step 2.

4 The error

4.1 Derivation

Sec. 3 gives every step; here is the one that matters (Step 7). The averaged Green's function is $\langle G\rangle = [E + \mu - H_0 - \langle V\rangle - \Sigma]^{-1}$. The renormalised parameters are defined by writing it as the Green's function of a clean block with barred parameters:

$$E + \mu_\sigma - h_\sigma(\boldsymbol{k}; m_\sigma) - \Sigma_\sigma = E + \bar\mu_\sigma - h_\sigma(\boldsymbol{k}; \bar m_\sigma). \tag{29}$$

The mass enters h_σ as $+m\,\tau_z$ (Eq. 1), the chemical potential as $-\mu\,\tau_0$. Identifying the τ_0 and τ_z components gives

$$\boxed{\operatorname{Re}\Sigma_\sigma = (\bar m_\sigma - m_\sigma)\,\tau_z - (\bar\mu_\sigma - \mu_\sigma)\,\tau_0} \tag{30}$$

The minus sign of Eq. (4) is right for μ and wrong for m. The τ_z component of g_σ is $(\bar m_\sigma - B_{\boldsymbol{k}})/\Delta_\sigma$ (Ref. [1], Eq. for g_z). The mass equation therefore reads

$$\boxed{\bar m_\sigma = m_\sigma + \frac{n_I J_I^2}{N^2}\sum_{\boldsymbol{k}}\Big[\frac{v_z}{\Delta_\sigma}(\bar m_\sigma - B_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar\sigma}}(\bar m_{\bar\sigma} - B_{\boldsymbol{k}})\Big]} \tag{31}$$

Eq. (5) for $\bar\mu_\sigma$ is unchanged.

4.2 One-line check

Treat a uniform, non-random term $\delta\,\tau_z$ as a self-energy: $\Sigma = \delta\,\tau_z$ exactly, and it shifts the mass to $m + \delta$. Eq. (4) gives $\bar m = m - \delta$.

4.3 It is not a sign convention

Some authors write $\langle G\rangle = [g^{-1} + \Sigma]^{-1}$ with $\Sigma = -\langle VgV\rangle$. A change of convention flips the τ_0 and τ_z components of Σ together. Eqs. (5) and (6) treat them differently: both carry a minus sign in front of the sum, while μ enters the Hamiltonian with a minus sign and m with a plus sign. No convention makes both equations right; the checks of Sec. 5 show which one is wrong.

4.4 Consistency with the perturbative result and with the topological Anderson insulator

At lowest order, Ref. [1] obtains

$$\operatorname{Re}\Sigma = \frac{n_I J_I^2}{8\pi}\,\frac{B\tau_z - D\tau_0}{B^2 - D^2}\Big[v_z \log\Big|\frac{(B^2 - D^2)k_*^4}{\Delta(E_F, 0)}\Big| + \dots\Big], \tag{32}$$

with a large positive logarithm when E_F is in the gap. With Eq. (30), the mass correction $\bar m - m = \operatorname{Re}\Sigma_{\tau_z}$ has the sign of B. It moves $\bar m$ away from zero on the topological side, $\bar m/B > 0$. This is

the renormalisation that explains the topological Anderson insulator [3, 4, 5], the mechanism the paper says it follows. With Eq. (4), the published equations push the masses towards the trivial side instead.

Physically, at second order the two couplings act alike on each block: τ_0 disorder (same-g) and τ_z disorder (opposite-g) give the same $\langle VgV\rangle$ once the terms odd in $\boldsymbol{k}$ are summed out, since $\tau_z(a+b\tau_z)\tau_z = a+b\tau_z$. At this order both act like the scalar disorder of the topological Anderson insulator. τ_z disorder is not scalar, though. At third order, $\tau_z g\,\tau_z g\,\tau_z = 2ab + (a^2+b^2)\tau_z$ for $g = a + b\tau_z$, whereas $g^2 = a^2 + b^2 + 2ab\,\tau_z$: the difference reappears with the third cumulant of the disorder (Sec. 5.2).

4.5 A second correction: the longitudinal weight

The longitudinal weight is printed as $v_z = \langle S_z^2\rangle - n_I^2 M_z^2 = (1-n_I^2)M_z^2 + (1-M_z)^2/3$. The prefactor n_I of Eq. (3) is already factored out of the second cumulant of the Bernoulli occupation, $n_I\langle S_z^2\rangle - n_I^2 M_z^2$. The weight should therefore be $v_z = (1-n_I)M_z^2 + (1-M_z)^2/3$. The printed value overestimates the longitudinal fluctuations, by a factor $1+n_I$ at $M_z = 1$, the case of this note. This factor is positive, so it changes the size of the fluctuation term and never its direction: the direction is set by the sign alone.

The size matters, though. With the corrected sign but the printed weight, the opposite-g spin-up inversion at $E_F = 0.02$ would fall at $J_I = 0.470$. With both corrections it lies at 0.410, and the exact masses put it at 0.400 (Sec. 5.2). The exact masses therefore decide v_z as well as the sign. The corrected curves below include both corrections.

5 Checks that do not use the derivation

5.1 Matrix SCBA

The SCBA can be solved without the reduction to four scalars. The self-energy is iterated as a 4×4 matrix, $\Sigma = \sum_t w_t V_t \langle G\rangle_{\mathrm{loc}} V_t$, where V_t are the exchange vertices with their second-cumulant weights and $\langle G\rangle_{\mathrm{loc}} = N^{-2}\sum_{\boldsymbol{k}}[E + \mathrm{i}\eta - H_0(\boldsymbol{k}) - \langle V\rangle - \Sigma]^{-1}$, on the same k grid. The reduced equations with the sign of Eq. (31) reproduce this matrix solution to 10^{-7}, for both couplings. With the sign of Eq. (6) they do not.

5.2 Exact renormalised masses

The disorder-averaged Green's function at $\boldsymbol{k} = 0$, at the energy $E + \mathrm{i}\eta$, defines the exact self-energy, to all orders in J_I:

$$\Sigma(E) = E + \mathrm{i}\eta - H_0(\mathbf{0}) - \langle G(\mathbf{0}, E+\mathrm{i}\eta)\rangle^{-1}. \tag{33}$$

In each block, $H_0(\mathbf{0}) = m\tau_z$ and $\langle G(\mathbf{0})\rangle$ is diagonal in the E/H basis, so

$$\bar{m} = m + \tfrac{1}{2}\,\mathrm{Re}(\Sigma_{EE} - \Sigma_{HH}), \qquad \bar{\mu} = -\tfrac{1}{2}\,\mathrm{Re}(\Sigma_{EE} + \Sigma_{HH}), \qquad \Gamma = -\tfrac{1}{2}\,\mathrm{Im}(\Sigma_{EE} + \Sigma_{HH}). \tag{34}$$

These include the mean field, like the SCBA's $\bar{m}_\sigma$ and $\bar{\mu}_\sigma$. The broadening Γ excludes η, so it stays near zero when no state lies at E.

We computed $\langle G(\mathbf{0})\rangle_{aa} = \langle\phi_a|(E+\mathrm{i}\eta - H)^{-1}|\phi_a\rangle$:

- ϕ_a is the normalised $\boldsymbol{k} = 0$ plane wave on orbital a;
- H is the Hamiltonian of a 160×160 periodic lattice with $\lfloor n_I \cdot 160^2 \rfloor$ impurities, and $\eta = 0.006$;
- the resolvent is expanded in Chebyshev polynomials of the rescaled Hamiltonian (Appendix A).

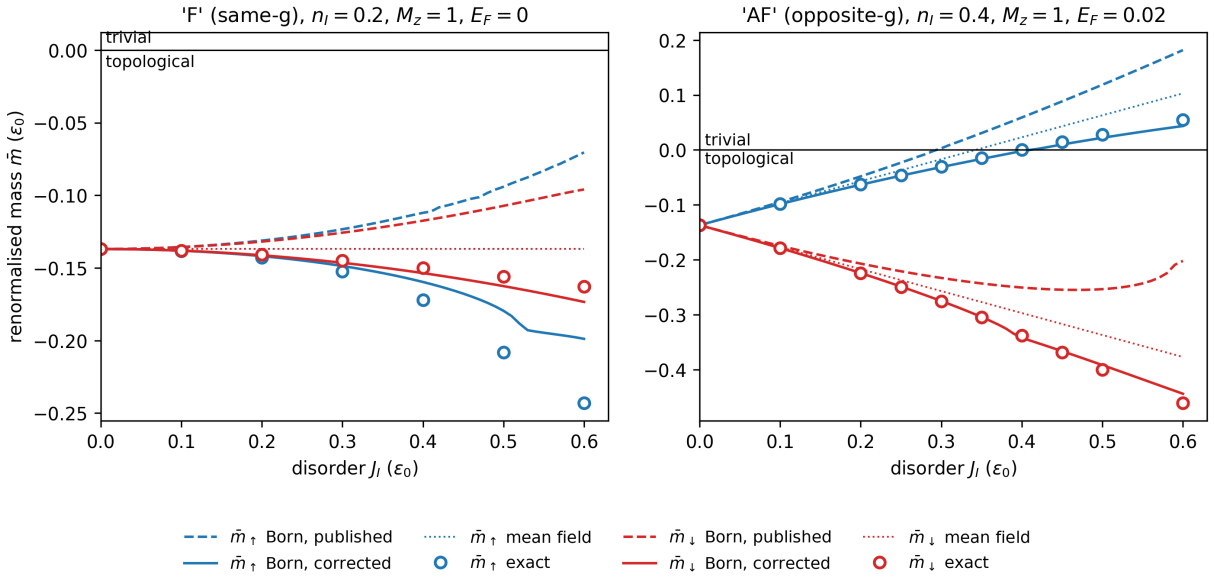


Figure 1: Renormalised masses: exact (circles), published SCBA (dashed), corrected SCBA (solid), mean field (dotted). A block is topological for $\bar{m} < 0$. Left: same-g, $n_I = 0.2$, $E_F = 0$. Right: opposite-g, $n_I = 0.4$, $E_F = 0.02$. The exact masses come from the disorder-averaged Green's function of a 160×160 periodic sample (no edges), 2 configurations, $\eta = 0.006$, with no Born approximation. The SCBA curves use the grid of the thesis program, 256×64 points of the Brillouin zone, with $\eta = 10^{-3}$.

Two independent samples agree to 10^{-3}. Fig. 1 and Table 1 compare the result with both versions of the SCBA.

- **Direction.** Measured from the mean field, the exact masses move towards the topological side, like the corrected SCBA and unlike the published one.
- **Weak disorder.** At same-g, $J_I = 0.2$, the corrected SCBA is within 10^{-3} of the exact masses. The published one is off by 0.01, in the opposite direction.
- **The chemical potential.** The exact $\bar{\mu}$ test the sign of Eq. (5), which the derivation keeps. At same-g, $E_F = 0$ and $J_I = 0.4$, the spin-up mean field is $\mu_\uparrow = -0.080$. The corrected SCBA gives $\bar{\mu}_\uparrow = -0.103$ and the exact value is -0.117. With the sign of Eq. (5) reversed, the SCBA would give -0.060, on the other side of the mean field. Spin down agrees: $+0.080$ for the mean field, $+0.067$ for the SCBA and $+0.070$ exact, against $+0.092$ with the reversed sign.
- **The inversion.** At opposite-g, the exact spin-up mass changes sign at $J_I = 0.40$, where the corrected SCBA puts it (0.41). The published SCBA puts it at 0.30. With the corrected sign but the printed v_z, it would be at 0.47 (Sec. 4.5).
- **Strong disorder.** The corrected SCBA remains close up to $J_I = 0.6$. At same-g the two blocks drift apart from it in opposite directions. This is the third cumulant of the binary disorder, which the SCBA leaves out (below).
- **The block that carries $G = 1$.** At same-g and $E_F = -0.05$, deep in the $G = 1$ plateau, the exact spin-down mass stays topological and its gap widens up to $J_I = 1.5$. The broadening there stays below 0.005: no state lies at E_F. Its local density of states at E_F, summed over $\boldsymbol{k}$, stays at 0.007–0.009 up to $J_I = 1.5$, the level of the η tail in the gap of the clean block (Appendix A). The published SCBA inverts this block at $J_I \approx 1.1$, which does not happen.
- **Spin up at strong disorder.** Table 2 follows the line $E_F = -0.06$, through the onset of the $G = 1$ plateau.

Table 1: Renormalised masses, exact and in the two versions of the SCBA.

case	J_I	exact	published	corrected
same-g, $\bar{m}_\uparrow$, $E_F = 0$	0.2	-0.1428	-0.1312	-0.1419
	0.4	-0.1720	-0.1121	-0.1597
	0.6	-0.2432	-0.0705	-0.1989
same-g, $\bar{m}_\downarrow$, $E_F = 0$	0.2	-0.1408	-0.1319	-0.1413
	0.4	-0.1500	-0.1175	-0.1536
	0.6	-0.1628	-0.0960	-0.1734
opposite-g, $\bar{m}_\uparrow$, $E_F = 0.02$	0.30	-0.0302	$+0.0029$	-0.0313
	0.35	-0.0147	$+0.0307$	-0.0164
	0.40	$+0.0002$	$+0.0591$	-0.0021
	0.45	$+0.0145$	$+0.0884$	$+0.0098$
opposite-g, $\bar{m}_\downarrow$, $E_F = 0.02$	0.40	-0.3377	-0.2506	-0.3422
same-g, $\bar{m}_\downarrow$, $E_F = -0.05$	0.8	-0.1790	-0.0605	-0.2084
	1.0	-0.1964	-0.0176	-0.2569
	1.2	-0.2150	$+0.0331$	-0.3129
	1.5	-0.2485	$+0.1235$	-0.3511

– For $0.4 \lesssim J_I \lesssim 0.8$, the spin-up states at E_F form an impurity band: the local density of states at E_F, summed over $\boldsymbol{k}$, is 6 to 12 times larger on the impurity sites than on the host sites.

– Near $J_I \approx 0.87$, $\langle G(\mathbf{0})\rangle_{HH}$ vanishes. Only the H component of $\langle G(\mathbf{0})\rangle^{-1}$ diverges: $\bar{\mu}_\uparrow$ and $\bar{m}_\uparrow$ diverge together, while $\mathrm{Re}\,\langle G(\mathbf{0})\rangle^{-1}_{EE} = E_F + \bar{\mu}_\uparrow - \bar{m}_\uparrow$ stays smooth, between -0.03 and -0.08. The broadening peaks at 1.2.

– $\bar{m}_\uparrow$ therefore changes sign through infinity, not through zero. This is neither a band inversion nor a topological transition, and the sign of $\bar{m}_\uparrow$ says nothing about the topology there.

– Between $J_I = 0.8$ and 1.0, the density of states at E_F drops by a factor 37: a gap opens. It is trivial, since $T_\uparrow = 0$ at every width, but the spin-up channel is lost before, from $J_I \approx 0.8$ (Fig. 5). Further on, at $E_F = -0.05$, the $\boldsymbol{k} = 0$ broadening falls to 0.01 at $J_I = 1.2$ and 0.006 at 1.5.

An impurity band crossing E_F calls for a T-matrix or CPA treatment, beyond any Born approximation. The published SCBA reaches a trivial spin-up block through a smooth inversion, at $J_I = 0.72$ on this line; the corrected one never does. Both predict $T_\uparrow = 0$ in this region, as observed. The spin-down block stays topological throughout, with no state at E_F.

Convergence. Fig. 2 repeats Fig. 1 with a 256×256 periodic sample, and with the SCBA on a 256×256 grid. Nothing moves:

- the exact masses change by at most 2×10^{-3} (at $J_I = 0.6$);
- the SCBA changes by at most 6×10^{-4}, and by at most 2×10^{-3} when η goes from 10^{-3} to 6×10^{-3};
- halving η to 0.003 changes the exact masses by at most 2×10^{-3}.

The remaining same-g offset is therefore not numerical.

Table 2: Exact $\boldsymbol{k} = 0$ parameters along $E_F = -0.06$ (same-g, $n_I = 0.2$), across the onset of the $G = 1$ plateau, with the published spin-up mass for comparison. $E_F + \bar{\mu}_\uparrow - \bar{m}_\uparrow$ is the real part of $\langle G(\mathbf{0})\rangle^{-1}_{EE}$ of the spin-up block.

J_I	$\bar{m}_\uparrow$	$\bar{\mu}_\uparrow$	$E_F + \bar{\mu}_\uparrow - \bar{m}_\uparrow$	$\Gamma_\uparrow$	published $\bar{m}_\uparrow$	$\bar{m}_\downarrow$	$\Gamma_\downarrow$
0.60	−0.270	−0.244	−0.034	0.10	−0.055	−0.164	0.0004
0.70	−0.273	−0.243	−0.030	0.21	−0.010	−0.171	0.0005
0.80	−0.569	−0.545	−0.036	0.37	+0.053	−0.179	0.0006
0.85	−1.212	−1.194	−0.042	1.24	+0.088	−0.183	0.0006
0.90	+1.176	+1.178	−0.058	0.38	+0.113	−0.188	0.0007
0.95	+0.552	+0.542	−0.070	0.08	+0.124	−0.192	0.0008
1.00	+0.369	+0.350	−0.080	0.04	+0.132	−0.197	0.0008

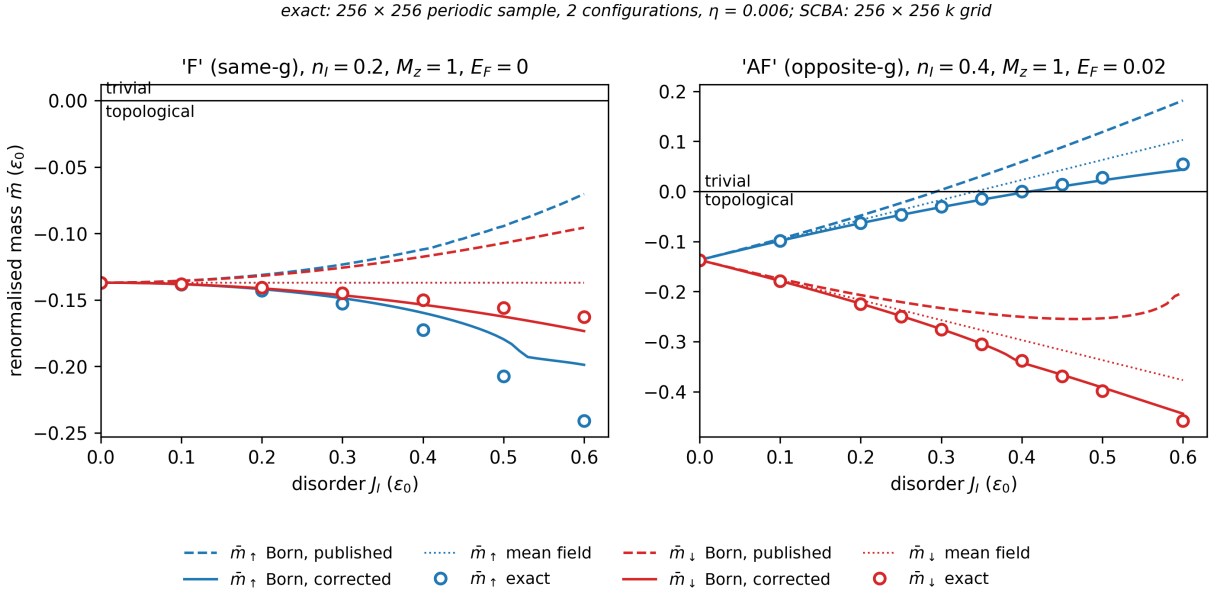


Figure 2: Fig. 1 with a 256×256 periodic sample for the exact masses (2 configurations, $\eta = 0.006$) and a 256×256 grid for the SCBA.

Where the same-g offset comes from. The SCBA keeps only the first two moments of the disorder (Step 2 of Sec. 3). The impurity potential, present or absent on each site, also has a third cumulant,

$$\langle (\delta U)^3 \rangle = \pm\, n_I(1 - n_I)(1 - 2n_I)\, J_I^3, \tag{35}$$

which enters the self-energy at third order.

- **Gaussian test.** We replaced the impurities by a Gaussian potential on every site, with the same mean and variance, so all higher cumulants vanish. The exact masses then fall onto the corrected SCBA, within 3×10^{-3} up to $J_I = 0.5$ (Table 3).
- **Opposite signs for the two spins.** At same-g, spin up sees $+J_I$ and spin down $-J_I$, so the third-order term changes sign with the spin. That is why the two blocks leave the SCBA in opposite directions.
- **Why opposite-g is so close.** Relative to the variance, the third cumulant scales as $(1 - 2n_I)J_I$: $0.6\, J_I$ at $n_I = 0.2$, but $0.2\, J_I$ at $n_I = 0.4$.

Table 3: Same-g, $n_I = 0.2$, $E_F = 0$: exact masses with the binary impurities and with a Gaussian potential of the same mean and variance (160×160 periodic sample, 2 configurations, $\eta = 0.006$), against the corrected SCBA.

	$\bar{m}_\uparrow$			$\bar{m}_\downarrow$		
J_I	binary	Gaussian	SCBA	binary	Gaussian	SCBA
0.3	−0.1525	−0.1491	−0.1487	−0.1448	−0.1466	−0.1465
0.4	−0.1720	−0.1628	−0.1597	−0.1500	−0.1541	−0.1536
0.5	−0.2082	−0.1777	−0.1795	−0.1560	−0.1635	−0.1625
0.6	−0.2432	−0.1899	−0.1989	−0.1628	−0.1759	−0.1734

Table 4: Exact spin-resolved transmissions and the renormalised mass of the relevant block in the two versions of the SCBA. Means over 8 configurations ($W = 64$, 128; ribbons of 400 sites) or 3 configurations ($W = 256$; disordered length 256). The $W = 64$ and 128 rows are a series of their own: the maps and cuts of Figs. 4–6 use other configurations (3 at $W = 128$), hence slightly different values. At $W = 128$ and $J_I = 0.35$, $T_\uparrow$ is 0.562 here and 0.52 there, within the spread of the three map configurations (0.50–0.57).

case	J_I	E_F	$T_\uparrow$	$T_\downarrow$	published	corrected
same-g, $W = 64$	1.2	−0.05	0.000	1.000	$\bar{m}_\downarrow = +0.033$ (trivial)	$\bar{m}_\downarrow = -0.313$
same-g, $W = 64$	1.3	−0.12	0.000	1.000	$\bar{m}_\downarrow = +0.080$ (trivial)	$\bar{m}_\downarrow = -0.314$
opposite-g, $W = 64$	0.30	0.02	0.840	1.000	$\bar{m}_\uparrow = +0.003$	$\bar{m}_\uparrow = -0.031$
opposite-g, $W = 64$	0.35	0.02	0.005	1.000	$\bar{m}_\uparrow = +0.031$ (trivial)	$\bar{m}_\uparrow = -0.016$
opposite-g, $W = 128$	0.30	0.02	0.997	1.000		
opposite-g, $W = 128$	0.35	0.02	0.562	1.000		
opposite-g, $W = 128$	0.40	0.02	0.012	1.000	$\bar{m}_\uparrow = +0.059$	$\bar{m}_\uparrow = -0.002$
opposite-g, $W = 256$	0.35	0.02	0.989	1.000	$\bar{m}_\uparrow = +0.031$ (trivial)	$\bar{m}_\uparrow = -0.016$
opposite-g, $W = 256$	0.40	0.02	0.339	1.000	$\bar{m}_\uparrow = +0.059$	$\bar{m}_\uparrow = -0.002$

5.3 Spin-resolved transport

With $M_z = 1$, the transmission splits exactly into $T = T_\uparrow + T_\downarrow$. We computed it on the ribbons of the paper's maps: 400×64 sites, $n_I \cdot 256 \cdot 64$ impurities in the central 256 columns, 8 configurations per point. Table 4 lists the points where the two versions disagree on the topology of a block.

- **Same-g.** The spin-down block carries a quantized channel, in 8 configurations out of 8, where the published masses make it trivial. A trivial block cannot carry a quantized chiral channel.
- **Opposite-g.** The spin-up channel vanishes earlier than the exact inversion, and later when the ribbon is wider. A trivial block would carry no channel at any width. This finite-width effect is the subject of Sec. 7.

6 Impact on the Born phase diagrams

Fig. 3 draws both versions of the Born lines over the published conductance maps. These are the data of Fig. 1 of Ref. [1], averaged over 101 configurations (opposite-g) and 50 configurations (same-g), in the colours of the thesis. Table 5 compares the $G = 1$ plateau with the points that each version predicts at $G = 1$. A block counts for one channel if it is topological and E_F lies in its gap ("localised" reading). Alternatively it counts if it is topological, whatever E_F ("topological" reading).

Coverage alone proves little: a prediction of $G = 1$ everywhere would cover the whole plateau.

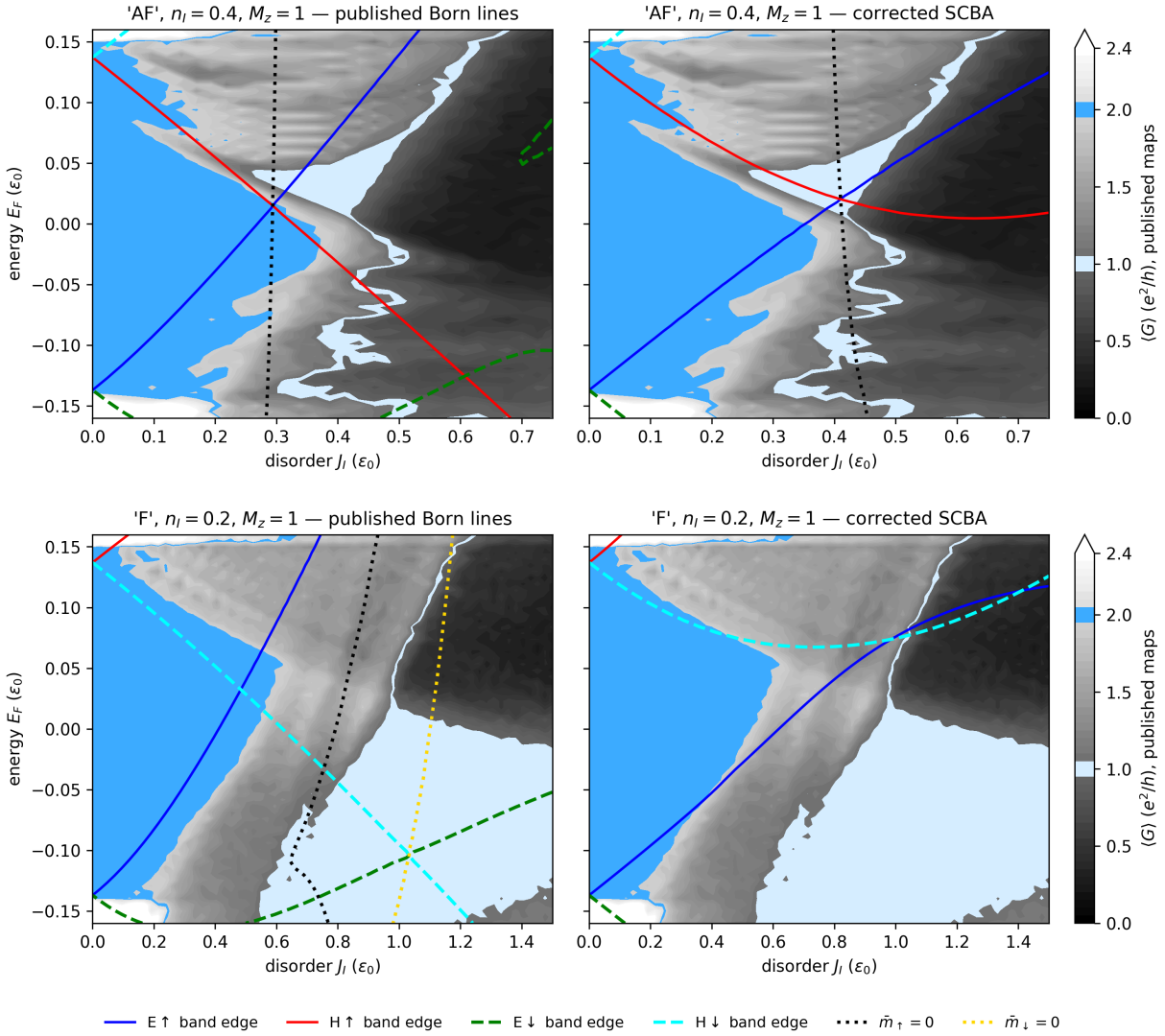


Figure 3: Born lines over the published conductance maps (Fig. 1 of Ref. [1]): published (left) and corrected SCBA (right). Light blue: $|\langle G \rangle - 1| < 0.05$; blue: $|\langle G \rangle - 2| < 0.05$. Solid lines: spin-up band edges; dashed: spin-down band edges; dotted: $\bar{m}_\sigma = 0$.

Table 5: The $G = 1$ plateau ($|\langle G\rangle - 1| < 0.05$) of the published maps against the points that the Born lines predict at $G = 1$. Coverage: share of the plateau predicted at $G = 1$. Precision: share of the points predicted at $G = 1$ that lie on the plateau. Predicting $G = 1$ everywhere would give a coverage of 100 % and a precision of 26 % (same-g) or 6 % (opposite-g). The corrected lines never make a same-g block trivial, so their topological reading predicts no $G = 1$ point there.

			localised reading		topological reading	
map		$G = 1$ points	published	corrected	published	corrected
same-g, $n_I = 0.2$	coverage	1065	20 %	99.9 %	39 %	0 %
	precision		17 %	35 %	51 %	–
opposite-g, $n_I = 0.4$	coverage	122	94 %	88 %	96 %	59 %
	precision		7 %	7 %	9 %	8 %

- **Same-g.** The corrected lines, in the localised reading, cover the plateau, and 35 % of their $G = 1$ points lie on it. The others lie on the crossover before the plateau, where E_F has left the spin-up gap but the spin-up states still conduct (Table 6). In the localised reading the published lines do worse than predicting $G = 1$ everywhere, with 17 %. In the topological reading they reach 51 %, but cover only 39 % of the plateau.
- **Opposite-g.** No version does much better than chance (7–9 % against 6 %). In the 64-site ribbons the $G = 1$ window is narrow, bounded by the width on the left and by the impurity states of the spin-down gap on the right (Secs. 7 and 8), whereas both versions predict $G = 1$ well beyond it.

Same-g.

- **Published lines.** Both masses change sign: $\bar{m}_\uparrow = 0$ at $J_I \approx 0.65$–0.95, then $\bar{m}_\downarrow = 0$ at $J_I \approx 1.0$–1.2. Beyond the second line no block is topological, so G should vanish. The published map shows $G = 1$ there, up to $J_I = 1.5$: 566 of its 1065 $G = 1$ points lie where the published masses make both blocks trivial. The shaded $G = 1$ region of the paper's Fig. 4 covers only a small part of the plateau.
- **Corrected lines.** No mass changes sign. They locate where the spin-up channel starts to fall, at the spin-up band edge, and keep spin down topological, as the exact masses do. The strong-disorder end of the spin-up behaviour (Sec. 5.2) is beyond them.

Table 6 follows the published map along lines of constant E_F:

- G leaves 2 at the corrected spin-up band edge.
- The plateau begins 0.2–0.4 further, along a line with the same slope.
- The published $\bar{m}_\uparrow = 0$ line is almost vertical ($J_I = 0.65$–0.83). It lies within 0.04 of the onset of the plateau for $-0.11 \lesssim E_F \lesssim -0.06$, and 0.08–0.2 away elsewhere.
- On the line $E_F = -0.06$, a trivial gap does open in the spin-up block at $J_I \approx 0.9$, once an impurity band has crossed E_F (Sec. 5.2). The paper's picture of a trivial spin-up block therefore holds at strong disorder, but not through a band inversion, and only after the spin-up channel is lost.

Opposite-g.

- **Published line.** $\bar{m}_\uparrow = 0$ at $J_I \approx 0.29$ lies at the left edge of the $G = 1$ region, as the paper describes.

Table 6: Same-g published map along lines of constant E_F: where $\langle G\rangle$ leaves 2 (< 1.95) and where the $G = 1$ plateau begins ($|\langle G\rangle - 1| < 0.05$), against the corrected spin-up band edge and the published $\bar{m}_\uparrow = 0$ line.

E_F	G leaves 2	plateau begins	corrected E↑ edge	published $\bar{m}_\uparrow = 0$
−0.109	0.24	0.62	0.14	0.65
−0.084	0.31	0.64	0.26	0.68
−0.058	0.38	0.76	0.38	0.72
−0.033	0.48	0.83	0.48	0.75
−0.008	0.52	0.93	0.59	0.78
+0.018	0.57	0.98	0.70	0.81

- **Corrected line.** It lies at $J_I \approx 0.41$–0.45, inside that region.

Sec. 7 shows that the left edge of the $G = 1$ region is not the inversion but a finite-width effect.

7 Why the published inversion line matched the antiferromagnetic map

7.1 Overlap of the edge states

An edge state of a BHZ block with mass m decays into the bulk over $\xi = 1/\lambda_-$. Here $\lambda_\pm^2$ are the roots of $(E_0 - D\lambda^2)^2 = (m + B\lambda^2)^2 - A^2\lambda^2$ at the edge-state energy $E_0 = -mD/B$ [6]. For $|m| \ll A^2|B|/[4(B^2 - D^2)] \approx 1.5$,

$$\xi \simeq \frac{A|B|}{|m|\sqrt{B^2 - D^2}} \simeq \frac{1.50}{|m|} \text{ sites.} \tag{36}$$

The exact root gives $\xi = 10.7$ sites for the clean mass (the approximation, 11.0), 50 sites for $|m| = 0.03$ and 100 sites for $|m| = 0.015$. In a ribbon of width W, the two counter-propagating spin-up edge states overlap once ξ becomes comparable with W. The disorder then backscatters between them across the ribbon, and the block's channel is lost while the block is still topological [6]. For the paper's $W = 64$, this happens at $|\bar{m}_\uparrow| \approx 0.015$–$0.03$ ($\xi \approx 50$–100 sites), i.e. $J_I \approx 0.30$–0.35 at $n_I = 0.4$ (Table 1). That is where the published map leaves $G = 2$, where the published $\bar{m}_\uparrow = 0$ line happens to lie.

7.2 Wider ribbons

Fig. 5 (left) follows both blocks along $E_F = 0.02$ for $W = 64$, 128 and 256. The disordered length is 256, with 8, 3 and 3 configurations.

- **The spin-up channel disappears later in wider ribbons.** $T_\uparrow = 1/2$ at $J_I \approx 0.32$ ($W = 64$), 0.355 ($W = 128$) and 0.39 ($W = 256$). The transition converges on the exact inversion, $J_I = 0.40$ (Table 1), and the channel is lost when ξ becomes comparable with W.
- **The published inversion is ruled out.** At $W = 256$ and $J_I = 0.35$, the spin-up block carries $T_\uparrow = 0.989$, the same within 0.002 in the three configurations. The published SCBA makes this block trivial there, with a gap of 0.06. A trivial block carries no channel at any width.
- **The spin-down channel also survives longer.** At $W = 256$, $T_\downarrow = 1.000$ up to $J_I = 0.475$, and it stays above 0.8 on average up to 0.6. At $W = 64$ it is already 0.89 at $J_I = 0.45$ and 0.60 at 0.5. Its loss in narrow ribbons comes from the impurity states that fill its gap (Sec. 8), which connect the two edges more easily when they are close.
- **The opposite-g $G = 1$ window moves and widens.** At $W = 256$ it starts at the exact inversion, $J_I \approx 0.40$, and extends beyond $J_I = 0.6$.

Fig. 5 (right) follows the same-g line $E_F = -0.06$.

- The spin-down block carries $T_\downarrow = 1.000$ at every J_I up to 1.5 and at every width, far beyond the published $\bar{m}_\downarrow = 0$ line.
- $T_\uparrow$ starts to fall at the corrected spin-up band edge ($J_I \approx 0.38$) at every width. It then crosses a diffusive region, where its value grows with W, and vanishes from $J_I \approx 0.8$ at every width.

Fig. 4 shows the full maps. The caroli maps at $W = 64$ reproduce the published ones. At $W = 128$:

- the same-g plateau is unchanged;
- the opposite-g $G = 2$ region extends to the corrected inversion, and the $G = 1$ region sits in the wedge beyond it, where the spin-up gap is trivial.

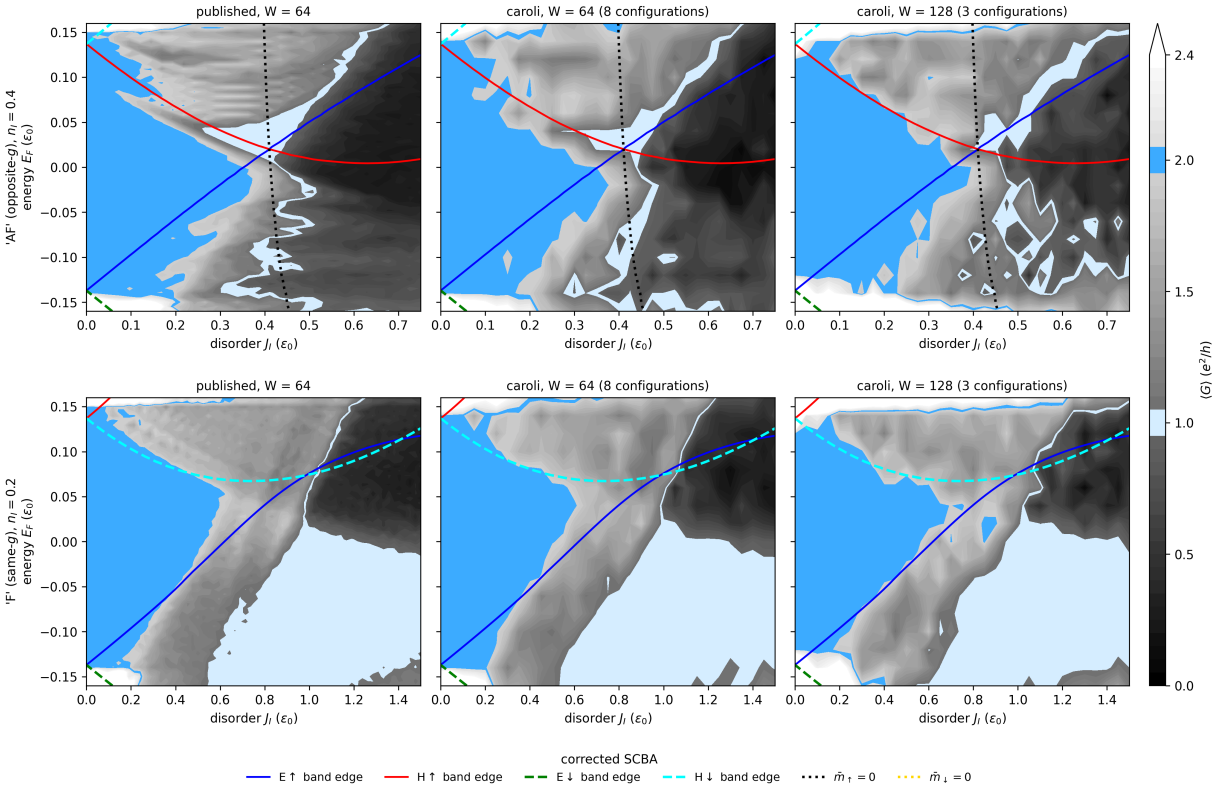


Figure 4: $\langle G \rangle$ maps in the colours of the thesis: published map ($W = 64$, Fortran, 101 and 50 configurations), and caroli on the same disordered length for $W = 64$ (8 configurations) and $W = 128$ (3 configurations). Lines: corrected SCBA.

7.3 The coincidence

The wrong sign moves the Born inversion from $J_I = 0.40$ to 0.30. The finite width moves the observed loss of the spin-up channel from 0.40 to about 0.32–0.35 at $W = 64$. The two shifts have the same direction and similar sizes, so the published line happened to fall on the edge of the published map. The same wrong sign predicts that both blocks become trivial at same-g, which the published map itself contradicts (Sec. 6).

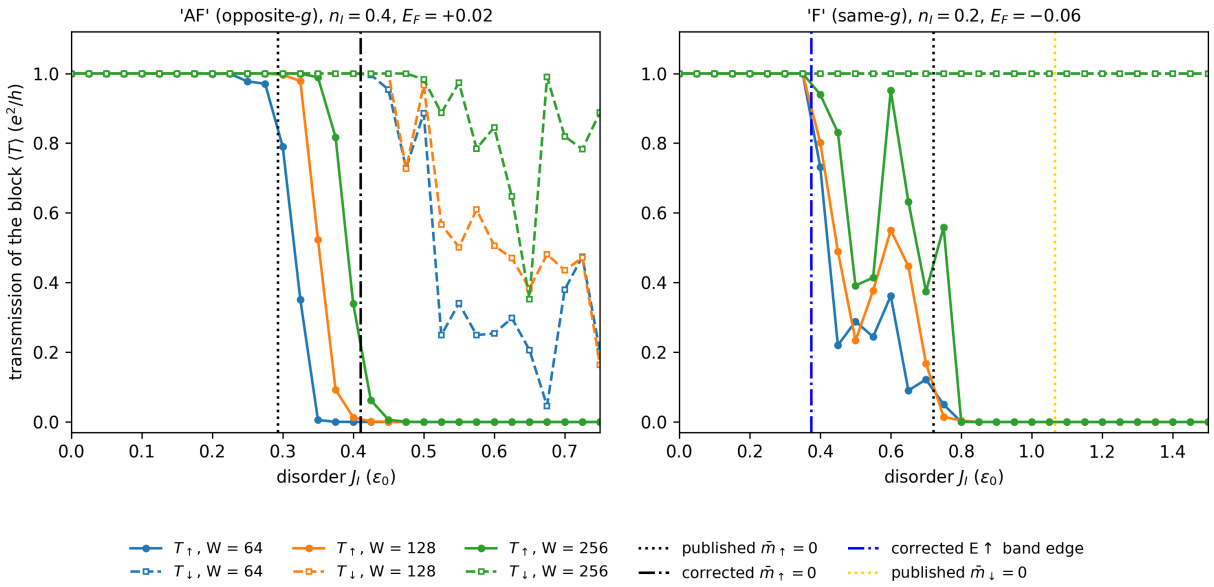


Figure 5: Transmissions of the two blocks along a line of constant E_F for $W = 64$, 128 and 256 (8, 3 and 3 configurations; disordered length 256). Left: opposite-g, $E_F = 0.02$, with the published and corrected spin-up inversions. Right: same-g, $E_F = -0.06$, with the corrected spin-up band edge and the published inversions of both blocks.

8 Origin and stability of the $G = 1$ phase

Fig. 6 resolves the conductance by spin. In each block, the corrected band edges of that block bound the region where it carries one channel.

Two independent Chern insulators. With $M_z = 1$, s_z is conserved and each block is a two-dimensional system without time-reversal symmetry (class A). Its two-terminal conductance is quantized to the Chern number of the occupied states whenever the states at E_F are localised. A block therefore contributes one channel as long as E_F lies above the delocalised states of its valence band and below those of its conduction band.

Same-g: how the phase appears.

- The mean exchange field acts as a spin-dependent chemical potential: the spin-up bands rise by $n_I J_I$ and the spin-down bands sink by $n_I J_I$.
- The fluctuations widen both gaps (Fig. 1).
- Near $E_F = 0$, the Fermi level enters the spin-up valence band as J_I grows: $T_\uparrow$ starts to fall at the corrected spin-up band edge.
- Once E_F lies below the delocalised spin-up states, the spin-up states at E_F are localised and carry no Chern number, so $T_\uparrow = 0$. The grey crossover of the map is the passage through those states.
- At stronger disorder ($J_I \gtrsim 0.9$ on $E_F = -0.06$), the impurity band that crossed E_F leaves it, and a trivial gap opens at E_F (Sec. 5.2). $T_\uparrow$ stays zero.
- The spin-down gap still contains E_F, so $T_\downarrow = 1$.
- The plateau ends only when E_F reaches the spin-down conduction band, at the top right of the map.

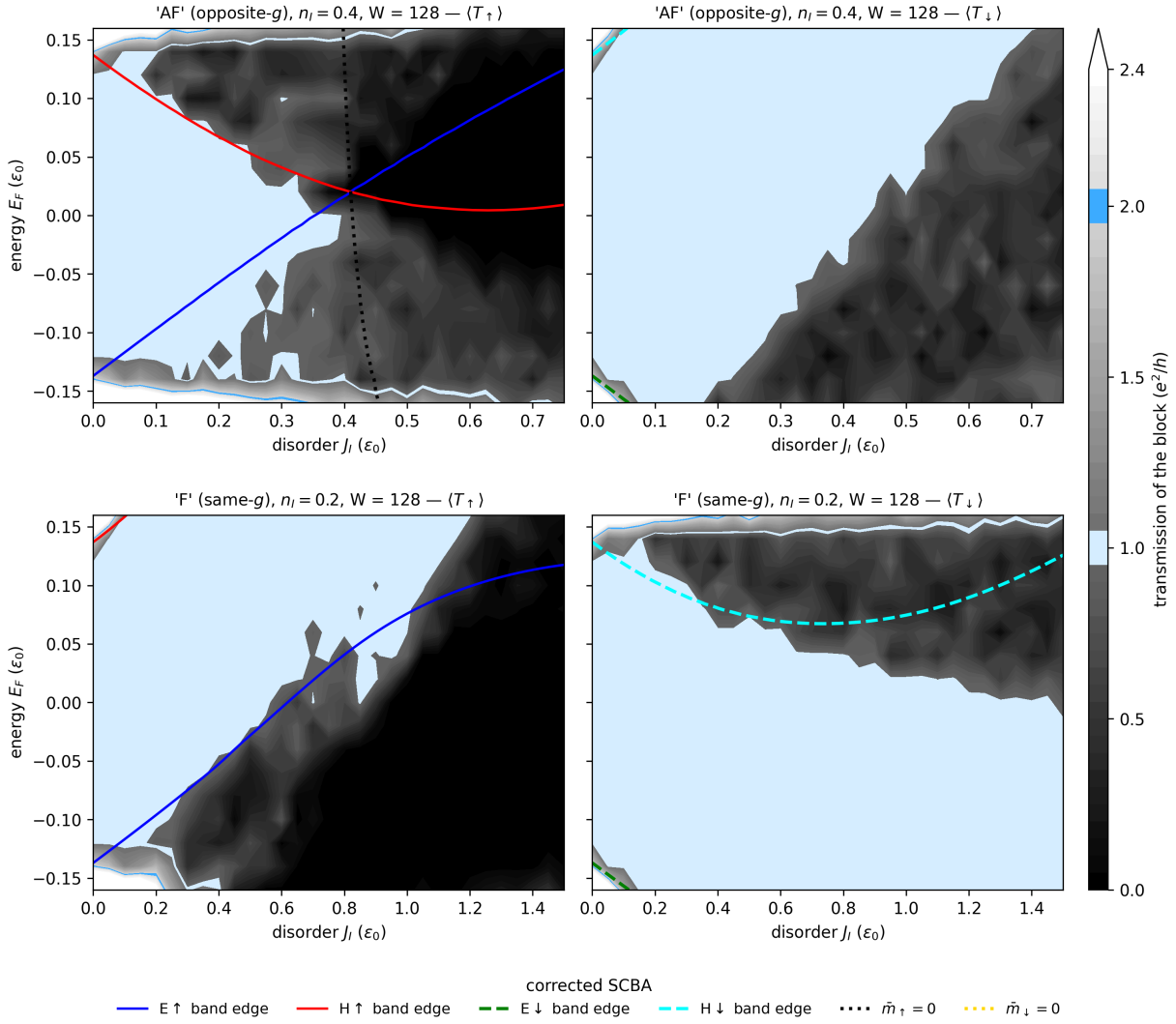


Figure 6: Spin-resolved transmissions $\langle T_\uparrow \rangle$ (left) and $\langle T_\downarrow \rangle$ (right), $W = 128$, 3 configurations, with the corrected band edges of each block, in the colours of Fig. 3. Light blue: the block carries one channel ($|\langle T_\sigma \rangle - 1| < 0.05$); black: none. Top: opposite-g, $n_I = 0.4$. Bottom: same-g, $n_I = 0.2$.

On the plateau of the $W = 64$ maps (8 configurations), $T_\downarrow$ is within 10^{-3} of 1 in 97 % of the single configurations, and $T_\uparrow < 10^{-3}$ in 92 %.

Why it is so stable.

- The $G = 1$ phase is a Chern insulator of total Chern number 1. Each edge carries a single chiral mode. There is no counter-propagating mode to scatter into, in either spin, so no disorder can backscatter it as long as the bulk states at E_F are localised.
- The $G = 2$ phase is different: each edge carries two counter-propagating modes, one per spin, and only s_z conservation keeps them apart.
- Tilting the moments shows the difference. With $M_z = 0.8$ the in-plane components couple the spins. On the thesis map at $M_z = 0.8$ (same-g, $n_I = 0.2$, 56 configurations), the $G = 2$ plateau shrinks from 23 % to 3.8 % of the map, while the $G = 1$ plateau keeps 13.9 % (against 26 % at $M_z = 1$).

Opposite-g.

- The mean field is a spin-dependent mass, so the spin-up block does change sign, at $J_I \simeq 0.40$ for $n_I = 0.4$. The fluctuations delay this inversion relative to the mean field (0.34).
- In Fig. 6, $T_\uparrow = 1$ fills the corrected topological gap of the spin-up block: the wedge between its two band edges, which closes at $J_I \approx 0.41$, $E_F \approx 0.02$. $T_\uparrow = 0$ fills the trivial gap that opens beyond.
- The $G = 1$ phase is the region where the spin-up gap is trivial while spin down still conducts.
- The spin-down block becomes more topological, but at $n_I = 0.4$ its gap fills with impurity states for $J_I \gtrsim 0.45$. The exact broadening at $E_F = 0.02$ rises from 0.001 at $J_I = 0.35$ to 0.013, 0.024 and 0.048 at $J_I = 0.45$, 0.5 and 0.6. At $W = 64$, $T_\downarrow$ drops to 0.89, 0.60 and 0.32.
- In the paper's 64-site ribbons, the $G = 1$ window is therefore narrow at this concentration. It widens in wider ribbons, where both channels survive longer (Sec. 7).

9 What changes for the paper

Unchanged.

- The numerical results: conductance maps (Fig. 1), local currents (Fig. 2) and local densities of states (Fig. 3).
- The main conclusion: a disorder-induced $G = 1$ Chern-insulator phase for both couplings, carried by the spin-down edge state and caused by the suppression of one spin channel through selective scattering and localisation.

Changed.

- The sign of the mass equations of the SCBA, and the weight v_z.
- Fig. 4 and the paragraphs that interpret it:
  - same-g: the spin-down mass never changes sign, whereas the published lines invert it at $J_I \approx 1.0$–1.2. The plateau begins where the spin-up states at E_F stop conducting, after E_F has entered the spin-up band, shifted by the exchange field. At stronger disorder an impurity band of the spin-up block crosses E_F, beyond the Born approximation, and a trivial gap opens only after the spin-up channel is lost;
  - opposite-g: the spin-up inversion lies at $J_I \simeq 0.40$ for $n_I = 0.4$, and the left edge of the $G = 1$ region in the maps is set by the ribbon width.

Draft addendum.

In the self-consistent Born analysis of Sec. IV B, the definition of the renormalised parameters should read $\mathrm{Re}\,\Sigma = \mathrm{diag}[(\bar{m}_\uparrow - m_\uparrow)\tau_z - (\bar{\mu}_\uparrow - \mu_\uparrow)\tau_0,\ (\bar{m}_\downarrow - m_\downarrow)\tau_z - (\bar{\mu}_\downarrow - \mu_\downarrow)\tau_0]$, so that the mass equations carry a plus sign in front of the sum. In addition, the longitudinal weight is $v_z = (1 - n_I)M_z^2 + (1 - M_z)^2/3$. The corrected Born phase diagrams are shown below.

With the correct sign, the fluctuations renormalise both masses towards the topological side, as in the topological Anderson insulator, and the same-g spin-down mass does not change sign. The $G = 1$ phase of the same-g case arises because the spin-dependent shift of the bands brings the Fermi level into the spin-up band, whose states are localised by the disorder. Meanwhile the spin-down block keeps a topological gap and its chiral edge state, as shown in Fig. 2. In the opposite-g case, the spin-up inversion occurs at $J_I \approx 0.40$ for $n_I = 0.4$. In the 64-site ribbons of Fig. 1 the $G = 1$ region starts earlier, near $J_I \approx 0.3$: close to the inversion, the spin-up edge states overlap across the ribbon and the spin-up channel is lost before the mass changes sign.

The numerical results and the conclusions of the paper are unchanged.

A Methods and reproducibility

The calculations use the Rust toolbox `caroli`, which reproduces the thesis programs on their own test cases, except the two independent checks at the end of this appendix. Each item below is an example, run with

```
cargo run --release -p caroli-validation --example <name>
```

- `born_lines`: published and corrected Born parameters, on the grids of the published maps.
- `born_self_energy`: exact renormalised masses (Sec. 5.2).
- `born_spin_channels`: spin-resolved transmissions of Table 4.
- `born_wide_maps`: spin-resolved maps and cuts for $W = 64, 128, 256$.

The figures come from the plotting scripts `plot_born_phase.py`, `plot_born_masses.py` and `plot_born_wide.py` in the same folder.

Chebyshev expansion of the resolvent. With $\tilde{H} = (H - c)/h$ rescaled into $(-1, 1)$, $\tilde{z} = (E + \mathrm{i}\eta - c)/h$, $s = \sqrt{\tilde{z}^2 - 1}$ and $t = \tilde{z} - s$ chosen with $|t| < 1$,

$$\langle\phi|(E + \mathrm{i}\eta - H)^{-1}|\phi\rangle = \frac{1}{h\,s}\Big[\mu_0 + 2\sum_{n\geq 1} t^n \mu_n\Big], \qquad \mu_n = \langle\phi|T_n(\tilde{H})|\phi\rangle. \tag{37}$$

The moments are obtained two per product with $\tilde{H}$, from $\mu_{2n} = 2\langle T_n|T_n\rangle - \mu_0$ and $\mu_{2n+1} = 2\langle T_n|T_{n+1}\rangle - \mu_1$. The series is truncated where $|t|^n < 10^{-10}$.

Spin-resolved transport. With $M_z = 1$ each block is a two-orbital BHZ model with its own disorder. Its transmission is computed by recursive Green's functions, with the lead self-energies from the propagating and evanescent modes. The blocks reproduce the four-orbital calculation to 10^{-12}.

Independent checks. Two results come from short scripts written outside `caroli` to check the note:

- the local densities of states at E_F of Sec. 5.2, $\rho = -\mathrm{Im}\,\mathrm{Tr}\,G_{\boldsymbol{rr}}/\pi$ averaged over 300 random sites, from direct sparse LU solves of $E + \mathrm{i}\eta - H$ on a 160×160 periodic sample ($\eta = 0.006$, one configuration). For the clean block they give 0.007 in the gap ($E = 0$) and 0.074 in the valence band ($E = -0.20$);
- the reduced SCBA with the sign of Eq. (5) reversed. With the printed sign, the same script reproduces the SCBA columns of Table 1 to 5×10^{-5}.

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